

LITTLE FALLS, MN, USA

WMO: 726578

Lat: 45.949N

Lon: 94.347W

Elev: 1122

StdP: 14.11

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04944

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.9	-12.6	-26.1	1.3	-17.4	-20.0	1.9	-9.9	19.7	18.1	17.9	15.5	2.0	290	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	89.7	72.7	85.6	69.9	81.9	67.4	75.3	83.9	73.0	81.6	71.0	79.0	10.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.5	125.4	80.0	70.3	116.3	78.1	68.2	107.9	75.8	39.5	84.0	37.4	82.0	35.5	78.9	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	16.7	14.7	DB	-25.9	93.8	6.4	3.5	-30.5	96.4	-34.3	98.4	-37.9	100.4	-42.5	103.0	(4)
(5)			WB	-26.3	78.7	5.8	3.0	-30.4	80.8	-33.9	82.6	-37.1	84.3	-41.4	86.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.7	12.4	16.3	29.2	42.6	55.1	65.0	69.5	66.7	59.0	45.7	31.2	17.8
	DBStd	22.28	13.42	13.22	12.10	9.72	8.39	6.92	5.91	5.81	8.55	8.94	10.63	12.39
	HDD50	4840	1165	944	649	258	42	1	0	0	18	195	568	1000
	HDD65	8555	1630	1364	1109	674	325	82	22	49	219	602	1014	1465
	CDD50	2166	0	0	5	36	200	450	606	517	288	60	4	0
	CDD65	403	0	0	0	1	17	81	163	101	38	2	0	0
	CDH74	4251	0	0	0	24	272	862	1674	1008	379	32	0	0
	CDH80	1199	0	0	0	4	73	240	530	261	88	3	0	0
Wind	WSAvg	5.7	5.5	5.7	6.1	6.5	6.5	5.6	4.9	4.6	5.4	6.2	6.0	5.4
Precipitation	PrecAvg	27.7	0.7	0.7	1.6	2.6	3.0	4.4	3.8	3.2	3.1	2.7	1.3	0.8
	PrecMax	35.4	2.4	1.7	4.0	7.0	6.3	7.2	8.7	7.2	6.1	6.5	4.3	1.8
	PrecMin	19.4	0.0	0.1	0.6	0.7	0.9	2.2	1.1	0.7	0.9	0.9	0.0	0.2
	PrecStd	3.8	0.6	0.5	0.9	1.5	1.4	1.5	1.7	1.7	1.4	1.5	1.1	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.4	49.8	66.0	77.3	87.8	91.6	93.2	90.9	88.1	78.8	63.8	45.4
		MCWB	37.2	42.8	54.9	56.7	66.2	73.4	75.6	74.7	69.8	62.4	51.5	39.4
	2%	DB	36.5	43.0	57.4	70.3	81.2	86.1	90.0	86.1	81.9	71.7	56.7	39.4
		MCWB	33.1	38.2	48.4	53.1	61.7	69.4	74.1	70.5	67.3	59.4	48.1	35.6
	5%	DB	33.6	37.4	51.7	64.3	75.8	81.8	85.9	82.1	79.0	64.4	51.5	36.2
		MCWB	31.8	34.5	45.8	49.6	58.8	66.2	71.3	68.3	64.9	54.9	44.3	33.9
	10%	DB	30.4	34.5	45.5	60.7	71.9	79.2	81.9	80.6	73.4	61.0	45.8	33.7
		MCWB	29.2	32.4	40.3	47.6	57.9	64.7	68.6	67.3	63.1	52.4	41.0	32.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.9	43.8	57.1	60.5	70.4	76.4	79.5	77.5	73.4	66.5	55.4	41.0
		MCDB	43.4	47.7	63.3	73.1	83.0	87.1	87.3	87.3	81.1	73.2	60.7	45.1
	2%	WB	33.4	38.8	50.2	55.1	65.5	72.9	76.4	73.8	70.7	61.0	49.5	36.6
		MCDB	35.4	42.2	55.8	66.1	75.6	82.2	85.3	81.9	78.7	67.8	55.7	39.5
	5%	WB	31.5	35.4	45.2	51.8	62.5	70.2	73.8	71.4	68.2	56.8	45.2	33.7
		MCDB	32.9	38.1	49.9	63.0	71.1	78.6	82.7	79.2	75.1	64.2	50.2	35.5
	10%	WB	29.3	32.6	40.1	48.4	59.5	67.4	71.6	69.3	65.7	53.3	41.3	31.8
		MCDB	30.5	34.4	45.4	58.8	68.8	75.2	79.6	75.7	72.2	58.9	46.3	32.9
Mean Daily Temperature Range	5% DB	MDBR	17.9	19.3	18.2	21.0	22.5	21.3	22.2	22.3	21.8	18.4	15.9	15.2
		MCDBR	17.9	19.2	25.2	30.4	30.3	25.6	25.2	25.6	26.7	26.3	24.4	18.4
		MCWBR	15.0	15.0	16.3	16.6	14.3	11.8	11.8	12.5	13.5	15.1	15.9	14.7
	5% WB	MCDBR	16.7	17.7	23.3	26.3	23.7	20.9	22.4	21.5	21.3	22.4	21.5	16.0
		MCWBR	14.4	14.0	15.8	16.1	13.5	11.5	12.1	12.3	12.5	14.6	15.2	13.4
Clear-Sky Solar Irradiance	taub	0.261	0.282	0.310	0.365	0.394	0.386	0.382	0.394	0.361	0.312	0.283	0.259	
	taud	2.388	2.324	2.375	2.305	2.281	2.343	2.377	2.347	2.415	2.539	2.506	2.423	
	Ebn at Noon	272	285	292	283	277	279	279	271	271	273	262	260	
	Edh at Noon	23	30	33	38	41	39	37	36	31	24	20	20	
All-Sky Solar Radiation	RadAvg	529	879	1211	1507	1708	1884	2008	1680	1274	796	501	367	
	RadStd	64	94	114	128	129	126	77	140	95	116	49	46	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (60 neighbors)	N/A	N/A	N/A	-2.02	-2.02	-2.56	N/A	N/A	N/A	-130
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page